

* Compound Interest:- Sometimes it so happens that the borrower and the lender agree to fix up a certain unit of time say yearly or half yearly or quarterly to settle the previous amount.

In such cases, the amount after first unit of time becomes the principal for the second unit, the amount after second unit becomes the principal for the third unit and so on.

After a specific period, the difference between the amount and the money borrowed is called the compound interest (C.I.) for that period.

Formulae

Let principal = P , Rate = $R\%$ per annum,
Time = n years.

I. When interest is compound annually :

$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^n$$

II. When interest is compounded Half-yearly:

$$\text{Amount} = P \left[1 + \frac{(R/2)}{100} \right]^{2n}.$$

III. When interest is compounded Quarterly:

$$\text{Amount} = P \left[1 + \frac{(R/4)}{100} \right]^{4n}$$

IV. When interest is compounded Annually but time is in fraction, say $3\frac{2}{5}$ years.

$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^3 \times \left(1 + \frac{\frac{2}{5}R}{100} \right).$$

V. When Rates are different for different years, say $R_1\%$, $R_2\%$, $R_3\%$ for 1st, 2nd and 3rd year respectively.

$$\text{Then, Amount} = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \left(1 + \frac{R_3}{100} \right).$$

VI. Present worth of Rs. x due n years hence is given by:

$$\text{Present worth} = \frac{x}{\left(1 + \frac{R}{100} \right)^n}$$

Ex:-1 Find compound interest on Rs. 7500 at 4% per annum for 2 years, compounded annually.

Soln Amount = Rs. $\left[7500 \times \left(1 + \frac{4}{100} \right)^2 \right] = \text{Rs. } \left(7500 \times \frac{26}{25} \times \frac{26}{25} \right)$
 $\Rightarrow$ Amount = 8112.

$\therefore$ C. I. = Rs. $(8112 - 7500) = \text{Rs. } 612$.

Ex:-2 Find compound interest on Rs. 8000 at 15% per annum for 2 years 4 months, compounded annually.

Soln Time = 2 years 4 months = $2\frac{4}{12}$ years = $2\frac{1}{3}$ years

Amount = Rs. $\left[8000 \times \left(1 + \frac{15}{100} \right)^2 \times \left(1 + \frac{\frac{1}{3} \times 15}{100} \right) \right]$
 $= \text{Rs. } \left(8000 \times \frac{23}{20} \times \frac{23}{20} \times \frac{21}{20} \right) = 11109$

$\therefore$ C. I. = Rs. $(11109 - 8000) = \text{Rs. } 3109$.

Ex:-3 Find the compound interest on Rs. 10,000 in 2 years at 4% per annum, the interest being compounded half yearly.

Soln Principal = 10,000; Rate = 2% per half year
 Time = 2 years = 4 half years.

$$\text{Amount} = \text{Rs.} \left[10,000 \times \left(1 + \frac{2}{100} \right)^4 \right] = \text{Rs.} \left(10,000 \times \frac{51}{50} \times \frac{51}{50} \times \frac{51}{50} \times \frac{51}{50} \right)$$

$$= \text{Rs.} 10824.32.$$

$$\therefore \text{C.I.} = \text{Rs.} (10824.32 - 10,000) = \text{Rs.} 824.32.$$

Ex:-4 Find the Compound interest on Rs. 16,000 at 20% per annum for 9 months compounded quarterly

Solⁿ Principal = Rs. 16,000 . Time = 9 months = 3 quarters;
Rate = 20 % per annum = 5 % per quarter.

$$\therefore \text{Amount} = \text{Rs.} \left[16,000 \times \left(1 + \frac{5}{100} \right)^3 \right]$$

$$= \text{Rs.} \left(16000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} \right) = \text{Rs.} 18522$$

$$\therefore \text{C.I.} = \text{Rs.} (18522 - 16000) = \text{Rs.} 2522.$$

Ex:-5 If the simple interest on a sum of money at 5% per annum for 3 years is Rs. 1200, find the compound interest on the same sum for the same period at the same rate.

Solⁿ Rate = 5% p.a , Time = 3 years , S.I. = 1200
Now Let Sum = P

$$\Rightarrow \text{S.I.} = \frac{P \times R \times T}{100} \Rightarrow 1200 = \frac{P \times 5 \times 3}{100}$$

$$\Rightarrow P = \frac{1200 \times 100}{5 \times 3} = 8000 \text{ Rs.}$$

$$\begin{aligned} \text{Amount} &= \text{Rs.} \left(8000 \times \left(1 + \frac{5}{100} \right)^3 \right) \\ &= \text{Rs.} \left(8000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} \right) = \text{Rs.} 9261 \end{aligned}$$

$$\text{C. I.} = \text{Rs.} (9261 - 8000) = \text{Rs.} 1261.$$

Ex-6 In what time will Rs. 1000 become Rs. 1331 at 10% per annum compounded annually?

Solⁿ Principal 1000, Amount 1331 Rs.

Rate 10% p.a.

Let the time be n years. Then,

$$\left[1000 \times \left(1 + \frac{10}{100} \right)^n \right] = 1331$$

$$\Rightarrow \left(\frac{11}{10} \right)^n = \left(\frac{1331}{1000} \right) = \left(\frac{11}{10} \right)^3$$

$$\Rightarrow n = 3 \text{ years.}$$

Ex-7 If Rs. 500 amounts to Rs. 583.20 in two years compounded annually, find the rate of interest per annum.

Solⁿ Principal = Rs. 500, Amount = Rs. 583.20, Time =

Let the rate be $R\%$ per annum. Then,

$$\left[500 \left(1 + \frac{R}{100} \right)^t \right] = 583.20$$

$$\text{or } \left(1 + \frac{R}{100} \right)^t = \frac{583.2}{500} = \frac{11664}{10000}$$

$$\therefore \left(1 + \frac{R}{100} \right)^t = \left(\frac{108}{100} \right)^2$$

$$\Rightarrow 1 + \frac{R}{100} = \frac{108}{100} \quad \text{or } R = 8$$

So Rate = 8% P.A.

Ex: 8 If the compound interest on a certain sum at $16\frac{2}{3}\%$ for 3 years is Rs. 1270, find the simple interest on the same sum at the same rate and for the same period.

Solⁿ Let the sum be Rs. x . Then,

$$C.I. = \left[x \times \left(1 + \frac{50}{3 \times 100} \right)^3 - x \right] = \left(\frac{343x}{216} - x \right)$$

$$C.I. = \frac{127x}{216} = 1270$$

$$\Rightarrow x = \frac{1270 \times 216}{127} = 2160.$$

Thus the sum is Rs. 2160.

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$$\therefore \text{S.I.} = \text{Rs.} \left(\frac{2160 \times 50}{3} \times 3 \times \frac{1}{100} \right) = \text{Rs.} 1080$$

Ex:-9 The difference between the compound interest and simple interest on a certain sum at 10% per annum for 2 years is Rs. 631. Find the sum.

Solⁿ Let the sum be Rs. x . Then,

$$\text{C.I.} = x \left(1 + \frac{10}{100} \right)^2 - x = \frac{21x}{100}$$

$$\text{S.I.} = \left(\frac{x \times 10 \times 2}{100} \right) = \frac{x}{5}$$

$$\therefore (\text{C.I.}) - \text{S.I.} = \left(\frac{21x}{100} - \frac{x}{5} \right) = \frac{x}{100}$$

$$\therefore \frac{x}{100} = 631 \Rightarrow x = 63100.$$

Hence, the sum is Rs. 63100.

Ex:-10 The difference between the compound interest and the simple interest on an amount of Rs. 18000 in 2 years was Rs. 405. What was the rate of interest?

Solⁿ Let the rate be $R\%$ p.a. Then,

$$\left[18000 \left(1 + \frac{R}{100} \right)^2 - 18000 \right] - \left(\frac{18000 \times R \times 2}{100} \right) = 405$$

$$\Rightarrow 18000 \left[\frac{(100+R)^2}{10,000} - 1 - \frac{2R}{100} \right] = 405$$

$$\Rightarrow 18000 \left[\frac{(100+R)^2 - 10,000 - 200R}{10000} \right] = 405$$

$$\Rightarrow \frac{9}{5} R^2 = 405 \quad \Rightarrow R^2 = \left(\frac{405 \times 5}{9} \right) = 225$$

$$\Rightarrow \cancel{202} R = 15$$

$\therefore$ Rate = 15%

Ex:-11 Divide Rs. 1301 between A and B, so that the amount of A after 7 years is equal to the amount of B after 9 years, the interest being compounded at 4% per annum.

Solⁿ Let the parts be Rs. x and Rs. $(1301-x)$.

$$x \left(1 + \frac{4}{100} \right)^7 = (1301-x) \left(1 + \frac{4}{100} \right)^9$$

$$\Rightarrow \frac{x}{(1301-x)} = \left(1 + \frac{4}{100} \right)^2 = \frac{26}{25} \times \frac{26}{25}$$

$$\Rightarrow 625x = 676(1301-x)$$

$$\Rightarrow 1301x = 676 \times 1301 \quad \Rightarrow x = 676$$

So the two parts are Rs. 676 and Rs. $(1301-676)$
i.e. Rs. 676 & Rs. 625.

Ex:-12 A certain sum amounts to Rs. 7350 in 2 years and to Rs. 8575 in 3 years. Find the sum and rate percent.

Soln S.I. on Rs. 7350 for 1 year. =

$$\text{Rs. } (8575 - 7350) = \text{Rs. } 1225.$$

$$\text{Rate} = \frac{100 \times 1225}{7350 \times 1} \% = 16 \frac{2}{3} \%$$

Let the sum be x , then

$$x \left(1 + \frac{50}{3 \times 100} \right)^3 = 7350$$

$$\Rightarrow x \times \frac{7}{6} \times \frac{7}{6} = 7350$$

$$\Rightarrow x = 7350 \times \frac{36}{49} = 5400.$$

$$\therefore \text{Sum} = 5400.$$

Ex:-13 A sum of money amounts to Rs. 6690 after 3 years and to Rs. 10,035 after 6 years on compound interest. Find the sum.

Soln Let the sum be Rs. P , Then.

$$P\left(1 + \frac{R}{100}\right)^3 = 6690 \quad \text{--- (1)}$$

$$\& P\left(1 + \frac{R}{100}\right)^6 = 10035 \quad \text{--- (2)}$$

On dividing we get -

$$\left(1 + \frac{R}{100}\right)^3 = \frac{10035}{6690} = \frac{3}{2}$$

Substituting this value in (1), we get -

$$P \times \frac{3}{2} = 6690 \Rightarrow P = 6690 \times \frac{2}{3} = 4460.$$

Hence, the sum is 4460.

Ex:-14 A sum of money doubles itself at compound interest in 15 years. In how many years will it become eight times?

$$\text{Soln} \quad P\left(1 + \frac{R}{100}\right)^{15} = 2P \Rightarrow \left(1 + \frac{R}{100}\right)^{15} = \frac{2P}{P} = 2 \quad \text{--- (1)}$$

$$\text{Let } P\left(1 + \frac{R}{100}\right)^n = 8P \Rightarrow \left(1 + \frac{R}{100}\right)^n = 8 = 2^3$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^n = \left\{\left(1 + \frac{R}{100}\right)^{15}\right\}^3 \quad (\text{using (1)})$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^n = \left(1 + \frac{R}{100}\right)^{45}$$

$$\Rightarrow n = 45.$$

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